

Unit - III

$$\textcircled{i) \sum P(x) = 1}$$

$$\Rightarrow 0.1 + k + 0.2 + 2k + 0.3 + 3k = 1$$

$$\Rightarrow 6k = 0.4$$

$$\Rightarrow k = \frac{1}{15}$$

x	-2	-1	0	1	2	3
$P(x)$	0.1	$\frac{1}{15}$	0.2	$\frac{2}{15}$	0.3	$\frac{1}{5}$

$$\textcircled{ii) } P(x < 2) = 1 - P(x \geq 2)$$

$$= 1 - [P(x=2) + P(x=3)] = 1 - [0.3 + \frac{1}{5}]$$

$$= \frac{1}{2} \text{ or } 0.5$$

$$\textcircled{iii) } P(-2 < x < 2) = P(x=-1) + P(x=0) + P(x=1)$$

$$= \frac{1}{15} + 0.2 + \frac{2}{15} = \frac{2}{5} \text{ or } 0.4$$

$$\textcircled{iv) } \text{Mean} = \mu = E(x) = \sum x_i \cdot P(x_i) = -2(0.1) - 1(\frac{1}{15}) + \frac{2}{15} + 2(0.3) + 3(\frac{1}{5}) = \frac{16}{15}$$

② wkt $\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^{\infty} kx^2 e^{-x} dx = 1$

$\Rightarrow k \int_0^{\infty} x^2 e^{-x} dx = 1$ Mean $= E(X) = \int_0^{\infty} \frac{x^3 e^{-x}}{2} dx$

$\Rightarrow k \Gamma(3) = 1$

$\Rightarrow k = \frac{1}{2} \quad \Gamma(4) = 3$

Variance $= \sigma^2 = E(X^2) - [E(X)]^2$

$= \int_0^{\infty} x^2 \cdot x^2 e^{-x} dx - 3^2 = \frac{1}{2} \Gamma(5) - 9$

$= 12 - 9 = 3$

③ Prop. that bomb strikes a target, $p = 0.2$

No. that a bomb misses the target, $q = 0.8$

$n = 6$, Let X = No. of bombs that strike.

$P(X=x) = {}^nC_x p^x q^{n-x} = {}^6C_x (0.2)^x (0.8)^{6-x}$

i) $P(X=2) = {}^6C_2 (0.2)^2 (0.8)^{6-2} = 0.2458$

ii) $P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$
 $= 1 - [{}^6C_0 (0.2)^0 (0.8)^6 + {}^6C_1 (0.2)^1 (0.8)^5]$
 $= 0.3446$

④ Given $\lambda = 3$, $n = 1000$

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-3} \cdot 3^x}{x!}$$

$$\begin{aligned} \text{i) } P(x > 3) &= 1 - P(x \leq 3) = 1 - [P(x=0) + P(x=1) + P(x=2) + P(x=3)] \\ &= 1 - \left[e^{-3} + e^{-3} \cdot 3 + \frac{e^{-3} \cdot 3^2}{2} + \frac{e^{-3} \cdot 3^3}{6} \right] \\ &= 0.3528 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(x \leq 3) &= P(x=0) + P(x=1) + P(x=2) + P(x=3) \\ &= 0.6472 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(x > 3) &= 1 - P(x < 3) = 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - \left[e^{-3} + 3e^{-3} + \frac{9e^{-3}}{2} \right] \\ &= 0.5768 \end{aligned}$$

⑤ $\mu_{\text{mean}} = 8 = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{8}$

$$f(x) = \begin{cases} \frac{1}{8} e^{-x/8}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{i) } P(0 < x < 1) &= \int_0^1 f(x) dx = \frac{1}{8} \int_0^1 e^{-x/8} dx = \left[-e^{-x/8} \right]_0^1 \\ &= -e^{-1/8} + e^0 = 1 - e^{-1/8} = 0.1813 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(-\infty < x < 10) &= \frac{1}{8} \int_{-\infty}^{10} e^{-x/8} dx = \left[-e^{-x/8} \right]_{-\infty}^{10} = -e^{-10/8} + 1 \\ &= 1 - e^{-1.25} = 0.8647 \end{aligned}$$

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$$(iii) P(x \leq 0) + P(x \geq 1) = 0 + \int_1^{\infty} \frac{1}{5} e^{-x/5} dx$$

$$= \left[-e^{-x/5} \right]_1^{\infty} = 0 + e^{-1/5} = 0.8187$$

⑥ Given Mean = 600 = $\frac{1}{\alpha}$

$$\alpha = \frac{1}{600} \quad x > 0$$

$$f(x) = \begin{cases} \frac{1}{600} e^{-x/600}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Let A be the sales & given that 9% of A is ₹ 81,

$$\Rightarrow A \times \frac{9}{100} = 81$$

$$\Rightarrow A = 900$$

$$\therefore P(x > 900) = \int_{900}^{\infty} f(x) dx = \int_{900}^{\infty} \frac{1}{600} e^{-x/600} dx$$

$$= \left[-e^{-x/600} \right]_{900}^{\infty} = 0 + e^{-3/2} = 0.2231$$

⑦ Mean = $\alpha\beta = 8$

Variance = $\alpha\beta^2 = 32$

$$\frac{\alpha\beta^2}{\alpha\beta} = \frac{32}{8}$$

$$\Rightarrow \alpha = 2, \beta = 4$$

$$\Gamma(\alpha, \beta, x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)}, \quad x \geq 0$$

otherwise

$$P(2, 4, x) = \begin{cases} \frac{x^2 - 1 - x/4}{16}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} x e^{-x/4} / 16, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} P(x \geq 10) &= \int_{10}^{\infty} f(x) dx = \int_{10}^{\infty} \frac{x e^{-x/4}}{16} dx = \frac{1}{16} \int_{10}^{\infty} x e^{-x/4} dx \\ &= \frac{1}{16} \left[\frac{x e^{-x/4}}{-1/4} - \frac{e^{-x/4}}{1/16} \right]_{10}^{\infty} = \frac{1}{16} \left[\frac{e^{-10/4}}{1/16} + \frac{10 e^{-10/4}}{1/4} \right] \\ &= 0.2873 \end{aligned}$$

⑧ Mean = $\alpha\beta = 8$; Variance = $\alpha\beta^2 = 21$

$$\frac{\alpha\beta^2}{\alpha\beta} = \beta = \frac{21}{8} ; \alpha = \frac{64}{21}$$

$$P(\alpha, \beta, x) = \begin{cases} \frac{x e^{-x/3}}{9\Gamma 2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\int_0^{\infty} \frac{x e^{-x/3}}{9} dx = \frac{1}{9} \left[-3x e^{-x/3} - 3 e^{-x/3} \right]_0^{\infty}$$

$$= \frac{1}{9} [0 + 18e^{-2} + 3e^{-2}] = \frac{21}{9} e^{-2} = 0.3158$$

$$\begin{aligned} \text{ii) } P(5 \leq x \leq 8) &= \frac{1}{9} \int_5^8 x e^{-x/3} dx = \frac{1}{9} \left[-3x e^{-x/3} - 3e^{-x/3} \right]_5^8 \\ &= \frac{1}{9} \left[(-24e^{-8/3} - 3e^{-8/3}) - (-15e^{-5/3} - 3e^{-5/3}) \right] \\ &= \frac{1}{9} \left[-27e^{-8/3} + 18e^{-5/3} \right] = 0.5862 \end{aligned}$$

⑨ $\mu = 2040, \sigma = 60 \quad z = \frac{x - \mu}{\sigma} = \frac{x - 2040}{60}$

$$\begin{aligned} \text{i) } P(x > 2150) &= P\left(z > \frac{2150 - 2040}{60}\right) = P(z > 1.83) \\ &= 0.5 - \phi(1.83) = 0.5 - 0.4664 = 0.0336 \end{aligned}$$

for 3000, $3000 \times 0.0336 = 101$

$$\begin{aligned} \text{ii) } P(x < 1950) &= P\left(z < \frac{1950 - 2040}{60}\right) = P(z < -1.5) \\ &= 0.5 - \phi(1.5) = 0.5 - 0.3531 = 0.1469 \end{aligned}$$

for 3000, $3000 \times 0.1469 = 441$

$$\begin{aligned} \text{iii) } P(1920 < x < 2160) &= P\left(\frac{1920 - 2040}{60} < z < \frac{2160 - 2040}{60}\right) \\ &= P(-2 < z < 2) = 2\phi(2) = 2 \times 0.4772 = 0.9544 \\ \text{for 3000, } &3000 \times 0.9544 = 2863 \end{aligned}$$

• ⑩ Same as ⑨ but for 2000 bulbs

- $\therefore$ i) $2000 \times 0.0336 \approx 67$ bulbs
 ii) $2000 \times 0.1496 \approx 299$ bulbs
 iii) $2000 \times 0.9544 \approx 1909$ bulbs

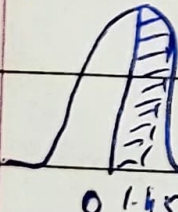
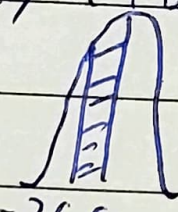
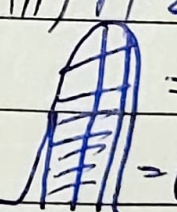
⑪ $\mu = 200, \sigma = 20, z = \frac{x - \mu}{\sigma} = \frac{x - 200}{20}$

a) $P(180 < x < 220) = P\left(\frac{180 - 200}{20} < z < \frac{220 - 200}{20}\right)$
 $= P(-1 < z < 1) = 2\phi(1) = 2 \times 0.3413 = 0.6826$

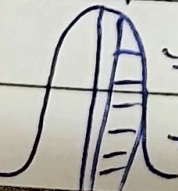
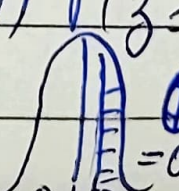
b) $P(x > 225) = P\left(z > \frac{225 - 200}{20}\right) = P(z > 1.25) = 0.5 - \phi(1.25)$
 $= 0.5 - 0.3944 = 0.1056$

c) $P(x < 150) = P\left(z < \frac{150 - 200}{20}\right) = P(z < -2.5) = 0.5 - \phi(2.5)$
 $= 0.5 - 0.4946 = 0.0054$

⑫ i) $P(0 \leq z \leq 1.45)$ ii) $P(-2.6 \leq z \leq 0)$ iii) $P(-3.4 \leq z \leq 2.65)$


 $= \phi(1.45)$
 $= 0.4265$

 $= \phi(2.6)$
 $= 0.4953$

 $= \phi(3.4) + \phi(2.65)$
 $= 0.4998 + 0.4960$
 $= 0.9878$

iv) $P(1.25 \leq z \leq 2.1)$ v) $P(z \geq 1.7)$


 $= \phi(2.1) - \phi(1.25)$
 $= 0.4821 - 0.3944 = 0.0877$

 $= 0.5 - \phi(1.7)$
 $= 0.5 - 0.4554 = 0.0446$